Machine Learning Lab 12

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Introduction

The aim of this lab was to classify biomedical abstract sentences into 5 categories (BACKGROUND, METHODS, RESULTS, OBJECTIVE, CONCLUSION) using Naive Bayes methods.

Tasks: (A) Implement Multinomial Naive Bayes from scratch, (B) Use sklearn MultinomialNB with TF-IDF and tuning, (C) Approximate Bayes Optimal Classifier using ensemble models.

Methodology

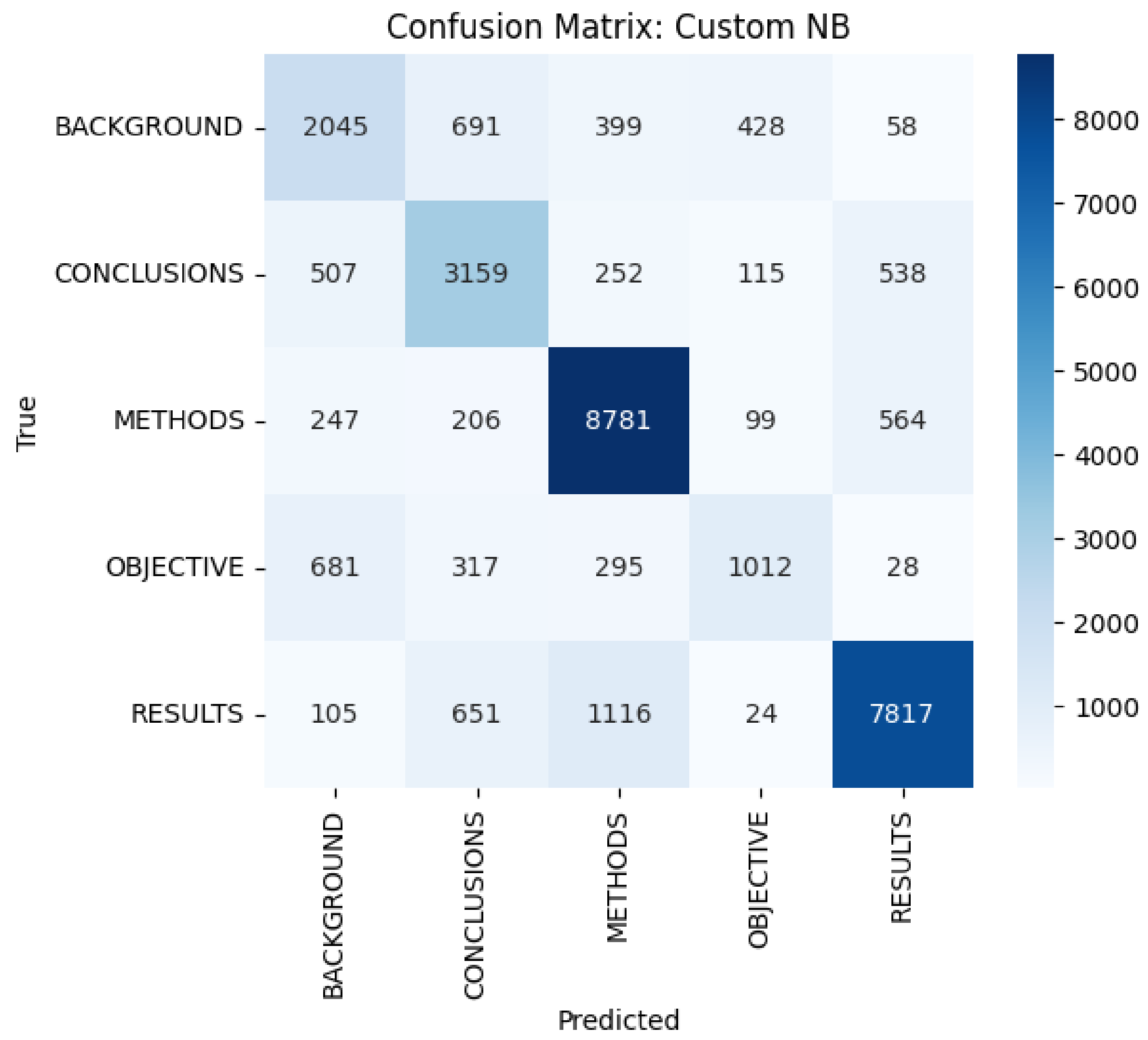
- Part A – Implemented custom Naive Bayes using CountVectorizer, Laplace smoothing, log priors/likelihoods, and argmax prediction.
- Part B – Used sklearn Pipeline (TfidfVectorizer + MultinomialNB) with GridSearchCV to tune n-grams and alpha.
- Part C – Created ensemble (MNB, Logistic Regression, Random Forest, Decision Tree, KNN) with posterior weights and soft voting to approximate BOC.

Results

Part A:

```
Fitting Count Vectorizer and transforming training data...
Vocabulary size: 301234
Transforming test data...

Training the Custom Naive Bayes Classifier (from scratch)...
Training complete.
```

```
=== Test Set Evaluation (Custom Count-Based Naive Bayes) ===
Accuracy: 0.7571
              precision    recall  f1-score   support

  BACKGROUND       0.57      0.56      0.57      3621
 CONCLUSIONS       0.63      0.69      0.66      4571
     METHODS       0.81      0.89      0.85      9897
   OBJECTIVE       0.60      0.43      0.50      2333
     RESULTS       0.87      0.80      0.84      9713

    accuracy                           0.76     30135
   macro avg       0.70      0.68      0.68     30135
weighted avg       0.76      0.76      0.75     30135

Macro-averaged F1 score: 0.6825
```

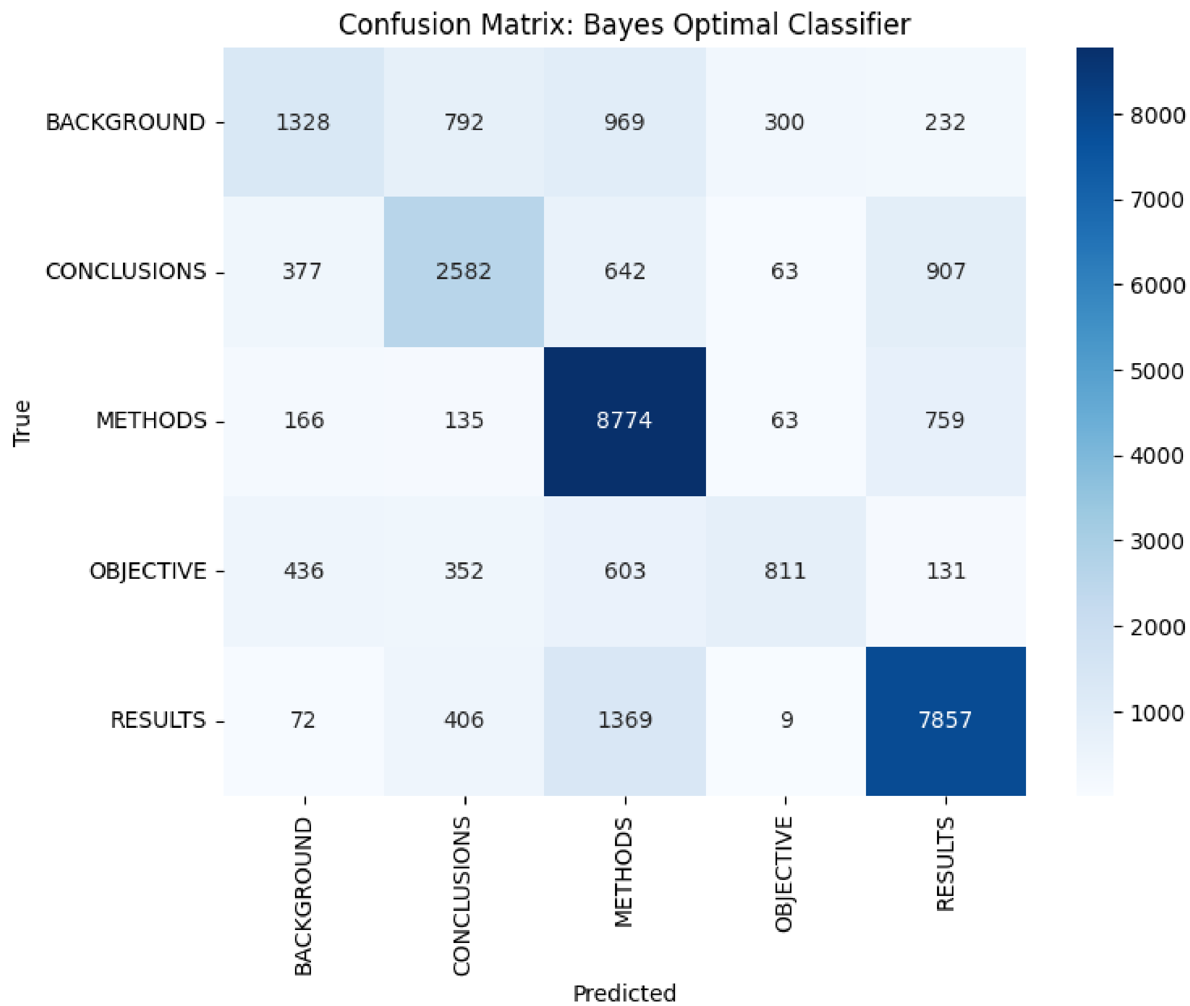


Part B:

```
Training initial Naive Bayes pipeline...
Training complete.

=== Test Set Evaluation (Initial Sklearn Model) ===
Accuracy: 0.6996
              precision    recall  f1-score   support

  BACKGROUND       0.61      0.37      0.46      3621
 CONCLUSIONS       0.61      0.55      0.57      4571
     METHODS       0.68      0.88      0.77      9897
   OBJECTIVE       0.72      0.09      0.16      2333
     RESULTS       0.77      0.85      0.81      9713

    accuracy                           0.70     30135
   macro avg       0.68      0.55      0.56     30135
weighted avg       0.69      0.70      0.67     30135

Macro-averaged F1 score: 0.5555

Starting Hyperparameter Tuning on Development Set...
Fitting 3 folds for each of 18 candidates, totalling 54 fits
Grid search complete.
Best params: {'nb__alpha': 0.1, 'tfidf__min_df': 5, 'tfidf__ngram_range': (1, 2)}
Best CV score: 0.6302879153606775
```

Part C:

```
Using dynamic sample size: 10173
Actual sampled training set size used: 10173

Training all base models...
Fitting NaiveBayes...
Fitting LogisticRegression...
Fitting RandomForest...
Fitting DecisionTree...
Fitting KNN...
All base models trained.
Posterior weights: [1.1692811745723865e-62, 1.0, 4.998616789348269e-81, 2.757203624177708e-301, 0.0]

Fitting the VotingClassifier (BOC approximation)...
Fitting complete.

Predicting on test set...

=== Final Evaluation: Bayes Optimal Classifier (Soft Voting) ===
Accuracy: 0.7085
              precision    recall  f1-score   support

  BACKGROUND       0.56      0.37      0.44      3621
 CONCLUSIONS       0.61      0.56      0.58      4571
     METHODS       0.71      0.89      0.79      9897
   OBJECTIVE       0.65      0.35      0.45      2333
     RESULTS       0.79      0.81      0.80      9713

    accuracy                           0.71     30135
   macro avg       0.66      0.59      0.61     30135
weighted avg       0.70      0.71      0.69     30135
```



Conclusion

1. Scratch model gave the lowest performance.
2. Tuned sklearn model improved results due to TF-IDF and optimized parameters.
3. BOC ensemble achieved the best and most balanced performance, combining multiple model strengths.